9. A customer presses vend without selecting an item

The customer presses the vend button before entering a complete item selection.

Objective	• Prevent the controller from vending an item when the customer has not made a valid selection.	
Results	• The controller blocks the vend and preserves any deposited money so the customer can select an item or cancel.	
Actors	• Customer, who understands the basic keypad and payment controls; vending machine controller.	
Related use cases	• Use Case 1: Buying an item. Use Case 3: Canceling a purchase. Use Case 4: Selecting an invalid location.	
Trigger	• The customer presses KEY3 before entering a complete, valid item selection.	
Stakeholders	• The machine owner and customers, who need the controller to prevent unintended purchases.	
Resources	• Programmed SoC Series V board, and Quartus.	
Assumptions	• A working setup with the seven-segment display, working money switches, working RAM set up for Items and the working keypad as well.	
Post-condition	• No item has been vended. The customer's deposited money remains credited, and the controller waits for a valid selection or cancellation.	
Frequency	• Whenever a customer presses the vend button before completing an item selection	
Mission Priority	• High. The controller must handle the vend button being pressed at any time without dispensing an unintended item.	
Safety Priority	• Should hold no safety concern	
Flow of events	**Step #**	**Action**
	1.	The customer moves SW3 from off to on to simulate inserting one dollar before selecting an item.
	2.	The controller records and displays a payment total of $1.00.

	3.	The customer presses KEY3 without entering an item location.
	4.	The controller checks the selection and determines that no valid item has been selected.
	5.	The technician uses the designated save control. The controller verifies that the location and price are valid, then updates location 11.
	6.	The controller provides a visible indication that an item must be selected. The customer can then make a selection or cancel.
Alternate Events	**Step #**	**Action**
	3A	If the customer presses KEY3 without selecting an item or depositing money, the controller blocks the vend and keeps the balance at $0.00.
	3B	If the customer enters only the first digit of a location before pressing KEY3, the controller treats the selection as incomplete and blocks the vend.
	6A	The customer selects location 11. The controller displays its $1.00 price. After the customer presses KEY3 again, the controller simulates vending the item, displays zero change, and clears the completed transaction.
	6B	The customer presses KEY1 to cancel. The controller displays a full refund of $1.00 and clears the transaction.
Hazard/ Result /Mitigation	• If case is used incorrectly, the entire system is likely incorrect or not working to specifications.	
Requirements from Use Case	• Block vending when no complete, valid item selection exists. • Preserve the deposited balance after a rejected vend attempt. • Provide a visible indication that the customer must select an item. • Allow the customer to complete a selection and press KEY3 again to purchase when payment is sufficient. • Allow KEY1 to cancel and return the full deposited amount before a successful vend.	